

Small-scale preparation, transfer, storage and cleanup of mixed laboratory chemicals for bench work.

REF
CAT-PDF-CHECK

ASSESSOR
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Research Lab

01 OVERVIEW

BUSINESS UNIT Research Lab	LOCATION Chemistry prep room	RISK ASSESSOR David Kemp	SOP REF NUMBER(S) SOP-CHEM-015
DATE OF ASSESSMENT 2026-06-10	DATE OF REVIEW 2028-06-10	PERSONS AT RISK Staff, Students	
RA TITLE Small-scale preparation, transfer, storage and cleanup of mixed laboratory chemicals for bench work.			

02 PROCESS STEPS & CHEMICALS

STEP	ACTIVITY	CHEMICALS	GHS SUMMARY	ENGINEERING	PPE	OTHER CONTROLS
Step 1	Prepare and transfer mixed chemicals Retrieve chemicals from storage, confirm current SDS, measure small working quantities in a fume hood, label working containers, return stock to segregated storage and collect wastes.	Acetone, Dichloromethane, Hydrochloric acid, Sodium hydroxide, Nitric acid, Sodium cyanide, Phenol, Toluene	     	Fume hood	Gloves, Goggles, Lab coat, Face shield	Use compatible secondary containment. Do not open incompatible chemicals together.

GHS SYMBOL KEY

 Flammable

 Oxidising

 Harmful / irritant

 Acute toxicity

 Health hazard

 Hazardous to environment

 Corrosive

See [Appendix A — Chemical Detail](#) for full hazard statements, WELs, and exposure information per chemical.

03 TASK HAZARDS

HAZARD	HOW HARM OCCURS	BEFORE CONTROLS	CONTROLS IN PLACE	AFTER CONTROLS	FURTHER ACTION
Glassware breakage and manual handling during chemical transfer	Cuts or dropped containers could cause injury and secondary chemical release.	L3 x S3 = 9	Use compatible trays, small containers, clear bench space and trained staff only.	L1 x S2 = 2	Brief staff before task.

04 CONTROLS

Elimination / substitution

Each chemical was reviewed. Remove any substance not required by the SOP before work starts.

Reduction

Use only the minimum working quantity at the bench; return stock containers immediately after dispensing.

Administrative controls

Only trained and authorised personnel may perform the task. Current SDS for all chemicals must be checked before work.

Monitoring / health

Air monitoring:

Review EH40/WEL information and SDS exposure limits.

Health surveillance:

Refer to Occupational Health where SDS, local procedure, sensitiser/CMR classification or potential exposure indicates surveillance.

Flammables Cabinet

Organic solvents and organic acids

Use for flammable organic solvents and flammable organic acids where SDS confirms this is appropriate.

Acetone

Toluene

Volatile Poisons / Chlorinated Solvents

Volatile poisons and chlorinated solvents

Requires secondary containment and suitable ventilation; keep away from incompatible spill contact and confirm SDS storage conditions.

Dichloromethane

Corrosives Cabinet

Non-oxidizing organic and mineral acids

Keep acids separate from bases; acids and bases should not share a cabinet.

Hydrochloric acid

Oxidizing acids in double containment

Double containment required; isolate from organic solvents/acids, bases and oxidizer cabinet contents.

Nitric acid

Corrosives Cabinet

Solid bases

Dry caustic bases such as hydroxide pellets/flakes; keep dry and separate from acids.

Sodium hydroxide

Toxins Cabinet

Non-volatile poisons - dry

Keep dry poisons separate from liquid poison spill paths.

Sodium cyanide

Phenol

06 EMERGENCY RESPONSE AND WASTE

SPILLS	Follow SDS Section 6 and the local spill procedure.
FIRST AID	Use SDS Section 4 as definitive first-aid instruction.
FIRE	Raise alarm and evacuate. Provide SDS and chemical inventory to responders.
WASTE HANDLING	Collect waste in compatible, labelled, closed containers under secondary containment.
OTHER	Emergency shower, eyewash, spill kit and SDS access point must be confirmed before starting.

07 BRIEFING & SIGN-OFF

NAME	DATE	SIGNATURE
Test user	2026-06-10	—

A1 APPENDIX A — CHEMICAL DETAIL

#	CHEMICAL	FORM / QTY	HAZARD STATEMENTS	WEL	STORAGE	EXPOSURE	GHS
1	Acetone CAS 67-64-1	Forms: liquid Mass/volume range: 250 mL	H225 H225; H319 H319; H336 H336	TWA: Check EH40/SDS STEL: — Source: Manual-EH40	Organic solvents and organic acids Check SDS sections 7 and 10 Source: GHS/PubChem + CAMEO Confidence: High GHS flammable H-code or pictogram. CAMEO groups: 19: Ketones H-codes: H225, H319, H336 GHS pictograms: flammable, harmful	Up to 30 minutes per use Weekly inhalation, skin, eye	 
2	Dichloromethane CAS 75-09-2	Forms: liquid Mass/volume range: 50 mL	H315 H315; H319 H319; H335 H335; H351 H351	TWA: Check EH40/SDS STEL: — Source: Manual-EH40	Volatile poisons and chlorinated solvents secondary containment Source: GHS/PubChem + CAMEO Confidence: High Volatile halogenated/chlorinated solvent signal. CAMEO groups: 17: Halogenated Organic Compounds H-codes: H315, H319, H335, H351 GHS pictograms: harmful, health-hazard	Up to 30 minutes per use Weekly inhalation, skin, eye	 
3	Hydrochloric acid CAS 7647-01-0	Forms: liquid Mass/volume range: 100 mL	H290 H290; H314 H314; H335 H335	TWA: Check EH40/SDS STEL: — Source: Manual-EH40	Non-oxidizing organic and mineral acids separate from bases Source: GHS/PubChem + CAMEO Confidence: High CAMEO acid reactive group. CAMEO groups: 1: Acids, Strong Non-oxidizing; 100: Water and Aqueous Solutions H-codes: H290, H314, H335 GHS pictograms: corrosive, harmful	Up to 30 minutes per use Weekly inhalation, skin, eye	 

#	CHEMICAL	FORM / QTY	HAZARD STATEMENTS	WEL	STORAGE	EXPOSURE	GHS
4	Sodium hydroxide CAS 1310-73-2	Forms: solid Mass/volume range: 100 g	H290 H290; H314 H314	TWA: Check EH40/SDS STEL: — Source: Manual-EH40	Solid bases separate from acids; keep dry; water-reactive - keep dry/review; incompatible absorbents: Cellulose-Based Absorbents, Mineral- Based & Clay-Based Absorbents Source: GHS/PubChem + CAMEO Confidence: High CAMEO base reactive group. CAMEO groups: 10: Bases, Strong H- codes: H290, H314 GHS pictograms: corrosive	Up to 30 minutes per use Weekly inhalation, skin, eye	
5	Nitric acid CAS 7697-37-2	Forms: liquid Mass/volume range: 25 mL	H272 H272; H290 H290; H314 H314	TWA: Check EH40/SDS STEL: — Source: Manual-EH40	Oxidizing acids in double containment double containment; isolate from organics, flammables and bases; water-reactive - keep dry/review; incompatible absorbents: Cellulose- Based Absorbents, Expanded Polymeric Absorbents Source: GHS/PubChem + CAMEO Confidence: High CAMEO strong oxidizing acid group. CAMEO groups: 2: Acids, Strong Oxidizing H-codes: H272, H290, H314 GHS pictograms: oxidising, corrosive	Up to 30 minutes per use Weekly inhalation, skin, eye	 
6	Sodium cyanide CAS 143-33-9	Forms: solid Mass/volume range: 10 g	H300 H300; H310 H310; H330 H330; H410 H410	TWA: Check EH40/SDS STEL: — Source: Manual-EH40	Non-volatile poisons - dry water-reactive - keep dry/review; hard separate from acids - HCN risk Source: GHS/PubChem + CAMEO Confidence: High CAMEO toxin reactive group. CAMEO groups: 11: Cyanides, Inorganic H-codes: H300, H310, H330, H410 GHS pictograms: toxic, environmental	Up to 30 minutes per use Weekly inhalation, skin, eye	 

#	CHEMICAL	FORM / QTY	HAZARD STATEMENTS	WEL	STORAGE	EXPOSURE	GHS
7	Phenol CAS 108-95-2	Forms: solid Mass/volume range: 25 g	H301 H301; H311 H311; H314 H314; H331 H331; H341 H341; H373 H373	TWA: Check EH40/SDS STEL: — Source: Manual-EH40	Non-volatile poisons - dry flammable - keep away from ignition sources and oxidizers; corrosive - use corrosion-resistant secondary containment Source: GHS/PubChem + CAMEO Confidence: High CAMEO toxin reactive group. CAMEO groups: 31: Phenols and Cresols H-codes: H301, H311, H314, H331, H341, H373 GHS pictograms: toxic, corrosive, health-hazard	Up to 30 minutes per use Weekly inhalation, skin, eye	
8	Toluene CAS 108-88-3	Forms: liquid Mass/volume range: 100 mL	H225 H225; H304 H304; H315 H315; H336 H336; H361 H361; H373 H373	TWA: Check EH40/SDS STEL: — Source: Manual-EH40	Organic solvents and organic acids retain toxic-label controls and secondary containment Source: GHS/PubChem + CAMEO Confidence: High GHS flammable H-code or pictogram. CAMEO groups: 16: Hydrocarbons, Aromatic H-codes: H225, H304, H315, H336, H361, H373 GHS pictograms: flammable, harmful, health-hazard	Up to 30 minutes per use Weekly inhalation, skin, eye	

B1 APPENDIX B — COSHH ESSENTIALS EXPLANATION & SCREENING

COSHH Essentials is an HSE control-banding screen. LabCAT presents it as substance-level screening: it estimates a control approach for each substance from health hazard group, quantity scale, and volatility or dustiness. It is not a legal approval by itself: a competent assessor must confirm the output against SDS information, exposure route, quantity, duration, WELs and local conditions.

GUIDANCE: COSHH ESSENTIALS REFERENCE TABLES

APPROACH 1	General ventilation and good practice may be sufficient only when confirmed by a competent assessor for the actual task.
APPROACH 2	Engineering control is indicated, normally LEV or equivalent capture/control.
APPROACH 3	Containment or enclosure is indicated where small breaches may occur.
APPROACH 4	Specialist advice is required; do not rely on the banded screening alone.

HAZARD GROUP	MEANING	EP BAND	MEANING
GROUP A	Lower health hazard band: H304, H315, H319, H336, EU66 and similar lower-toxicity effects.	EP1	Lowest exposure potential: small amounts with low/medium dustiness, or low-volatility millilitre-scale liquids.
GROUP B	Harmful / STOT category 2 band: H302, H312, H332, H371.	EP2	Low-to-moderate exposure potential: grams of high-dust solids, kg/tonne of low-dust solids, or millilitres of medium/high-volatility liquids.
GROUP C	Toxic, corrosive, serious eye damage, sensitising skin, respiratory irritation or STOT category 1/2: H301, H311, H314, H317, H318, H331, H335, H370, H373, EU71.	EP3	Moderate-to-high exposure potential: litre-scale liquid work or kilogram-scale medium/high-dust solid work.
GROUP D	Fatal acute toxicity, suspected carcinogen / reproductive toxicant, lactation hazard or repeated-exposure organ damage: H300, H310, H330, H351, H360, H361, H362, H372.	EP4	Highest exposure potential: cubic-metre-scale high-volatility liquids or tonne-scale medium/high-dust solids.
GROUP E	Respiratory sensitiser, mutagen, carcinogen category 1 / suspected mutagen or toxic by eye contact: H334, H340, H341, H350, EU70. Specialist advice is required (Approach 4).		

PREDICTED EXPOSURE RANGES BY EP BAND AND CONTROL APPROACH

This reference table shows the predicted airborne exposure ranges associated with each exposure predictor (EP) band and control approach. It helps explain why COSHH Essentials may move from general ventilation to engineering control or containment as the expected dust or vapour exposure increases. It is background guidance only: LabCAT still bases the recommendation for each substance on hazard group, quantity scale and volatility or dustiness, and the assessor must confirm the final controls against the task and SDS.

PREDICTED EXPOSURES FOR DUST IN AIR (MG/M3)			
EP BAND	CONTROL APPROACH 1	CONTROL APPROACH 2	CONTROL APPROACH 3
EP1 SOLID	0.01 to 0.1	0.001 to 0.01	<0.001
EP2 SOLID	0.1 to 1	0.01 to 0.1	0.001 to 0.01
EP3 SOLID	1 to 10	0.1 to 1	0.01 to 0.1
EP4 SOLID	>10	1 to 10	0.1 to 1

PREDICTED EXPOSURES FOR VAPOUR IN AIR (PPM)			
EP BAND	CONTROL APPROACH 1	CONTROL APPROACH 2	CONTROL APPROACH 3
EP1 LIQUID	<5	<0.5	<0.05
EP2 LIQUID	5 to 50	0.5 to 5	0.05 to 0.5
EP3 LIQUID	50 to 500	5 to 50	0.5 to 5
EP4 LIQUID	>500	5 to 500	0.5 to 5

RECOMMENDATIONS: SUBSTANCE-LEVEL SCREENING OUTPUT

The table below is LabCAT's substance-level COSHH Essentials screening result for this assessment. Each substance has its own approach; the highest approach present is highlighted for assessor review, but task-specific controls must still be confirmed.

SUBSTANCE	GROUP	H-CODES	SCALE	BAND	EP	APPROACH
Acetone	A	H319, H336	small	high (volatility)	EP2	Approach 1 — General ventilation
Dichloromethane	D	H351	small	high (volatility)	EP2	Approach 3 — Containment / enclosure
Hydrochloric acid	C	H314, H335	small	medium (volatility)	EP2	Approach 2 — Engineering controls (LEV)
Sodium hydroxide	C	H314	small	medium (dustiness)	EP1	Approach 1 — General ventilation
Nitric acid	C	H314	small	medium (volatility)	EP2	Approach 2 — Engineering controls (LEV)

SUBSTANCE	GROUP	H-CODES	SCALE	BAND	EP	APPROACH
Sodium cyanide	D	H300, H310, H330	small	high (dustiness)	EP2	Approach 3 — Containment / enclosure
Phenol	E	H341	small	medium (dustiness)	EP1	Approach 4 — Specialist advice required
Toluene	D	H361	small	medium (volatility)	EP2	Approach 3 — Containment / enclosure